# Data Structures and Algorithms -3 (25CS2103E)

## News and Social Media Analytics Engine


Team Members

Name: J Pragathi  Rollno. 2520090216

Name: J Rakshitha Rollno. 2520090214

Under the Esteemed Guidance of

Dr K Chandhusha

CSIT

# Contents

- Abstract
- Introduction
- Literature survey
- Problem Statement and Objectives
- Hardware and Software requirements
- Results
- Conclusion
- Future Scope

# Abstract

The News and Social Media Analytics Engine is a web-based application that demonstrates how data structures and algorithms can help organize and analyze news and social media posts. It provides features such as searching posts, detecting duplicate content, ranking posts by priority, grouping related stories, moderation, and relationship visualization. The frontend is developed using HTML, CSS, and JavaScript, while Java implements the processing modules. Node.js and Express connect the frontend with the Java backend. The current prototype uses a sample CSV dataset to demonstrate the working of different data structures.

## Introduction

- News and social media platforms generate large amounts of information.
- Finding relevant posts and identifying repeated content can be difficult.
- Data structures and algorithms help organize and process information efficiently.
- Our project combines multiple DSA concepts in one web application.

# Literature Survey

**Existing Systems and References**

**1. Google News**

Organizes news articles from multiple publishers.

Helps users explore news by topics and events.

**2. GDELT Project**

Collects and organizes global news information.

Supports research and analysis of news events.

**3. NewsAPI**

Provides interfaces for searching news articles and headlines.

Can be considered for future live-news integration.

**4. Java Collections Framework**

Provides data structures for storing and processing information.

Supports concepts such as queues and priority queues.

**Our Project:** Demonstrates similar content-organization concepts using a local sample dataset and custom DSA modules.

# Problem Statement and Objectives

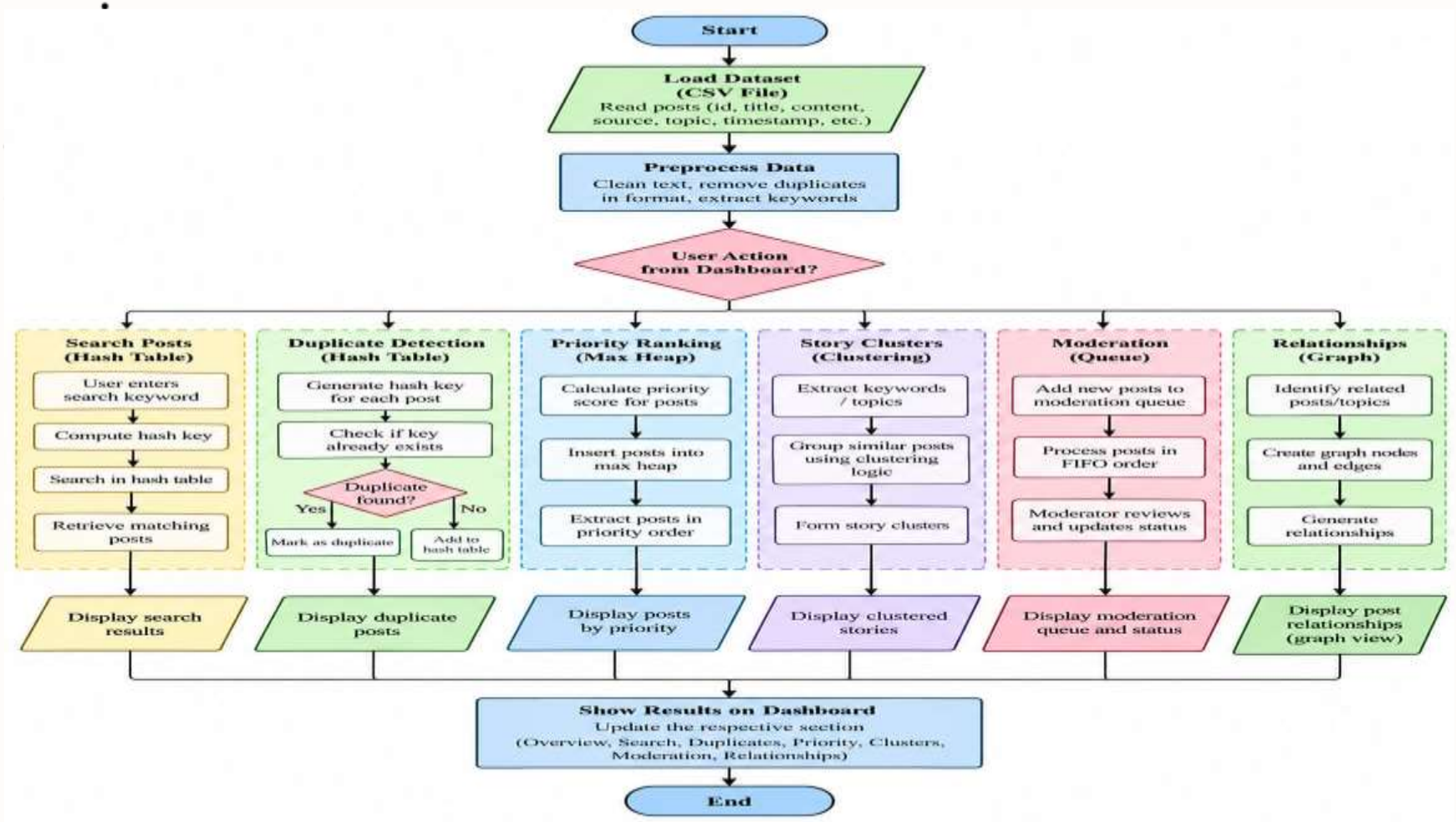
**Problem Statement**

Searching, organizing, and reviewing large collections of news and social media posts can be difficult when duplicate or related content is not systematically identified.

**Objectives**

- Search for specific posts.
- Identify duplicate content.
- Rank posts according to priority.
- Group related posts into story clusters.
- Support moderation and review.
- Display relationships between posts and topics.
- Demonstrate the practical use of Data Structures and Algorithms.

# Flow Chart


Start
Load Dataset
(CSV File)
Read posts (id, title, content, source, topic, timestamp, etc.)
Preprocess Data
Clean text, remove duplicates in format, extract keywords
User Action from Dashboard?
Search Posts (Hash Table)
User enters search keyword
Compute hash key
Search in hash table
Retrieve matching posts
Display search results
Duplicate Detection (Hash Table)
Generate hash key for each post
Check if key already exists
Duplicate found?
Yes
No
Mark as duplicate
Add to hash table
Display duplicate posts
Priority Ranking (Max Heap)
Calculate priority score for posts
Insert posts into max heap
Extract posts in priority order
Display posts by priority
Story Clusters (Clustering)
Extract keywords / topics
Group similar posts using clustering logic
Form story clusters
Display clustered stories
Moderation (Queue)
Add new posts to moderation queue
Process posts in FIFO order
Moderator reviews and updates status
Display moderation queue and status
Relationships (Graph)
Identify related posts/topics
Create graph nodes and edges
Generate relationships
Display post relationships (graph view)
Show Results on Dashboard
Update the respective section
(Overview, Search, Duplicates, Priority, Clusters, Moderation, Relationships)
End

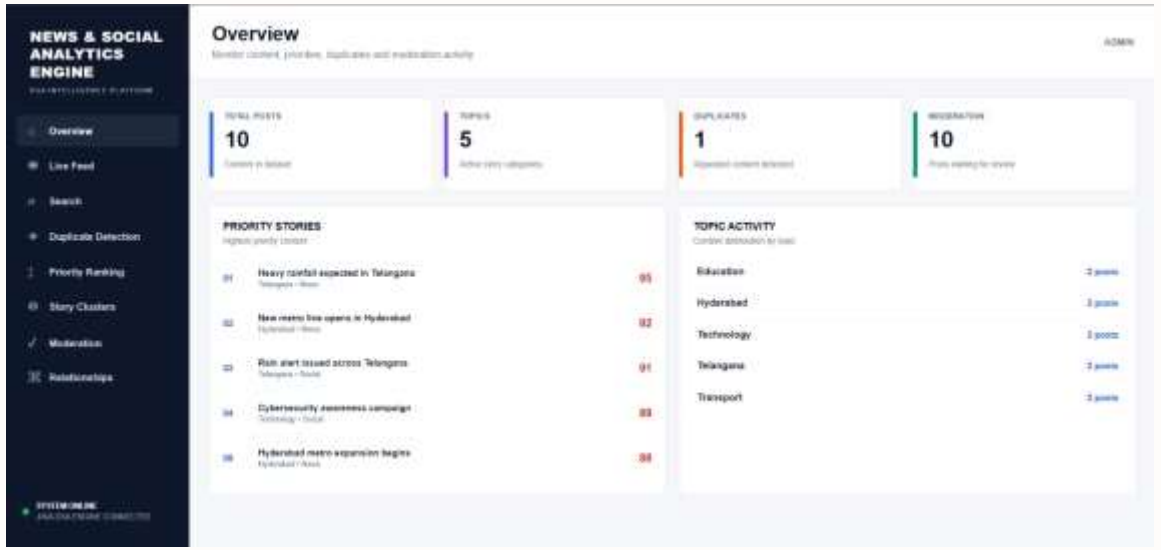
# Results

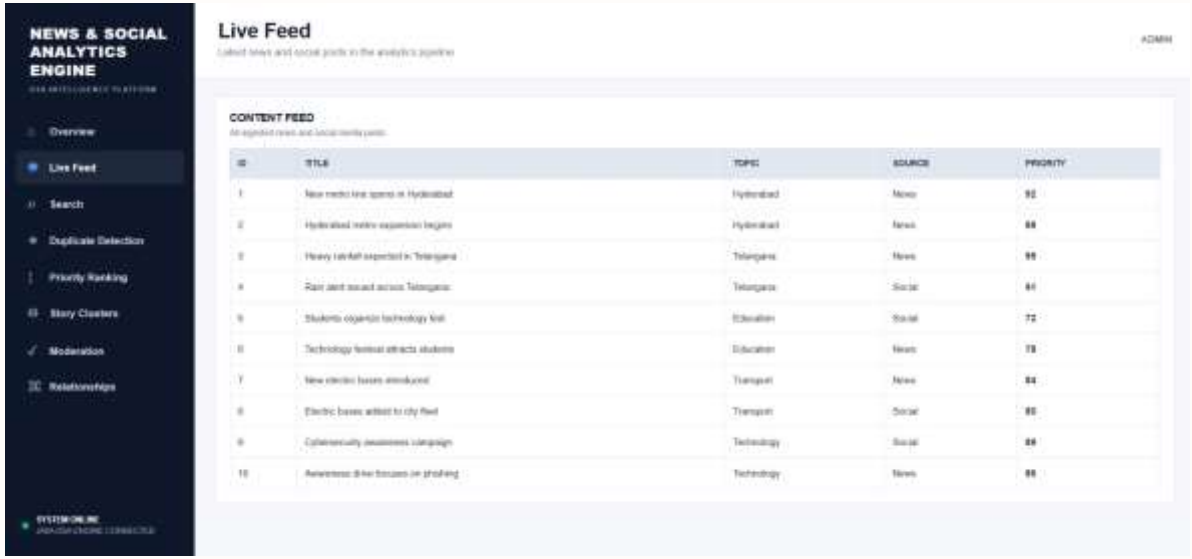
## Results of the Working Prototype

- **Overview:** Displays a summary of the available posts.
- **Live Feed:** Shows posts from the sample dataset.
- **Search:** Finds posts matching a search term.
- **Duplicate Detection:** Identifies repeated content according to the implemented rule.
- **Priority Ranking:** Displays posts according to priority.
- **Story Clusters:** Groups posts using the implemented grouping logic.
- **Moderation:** Displays the review workflow.
- **Relationships:** Shows connections between posts or topics.

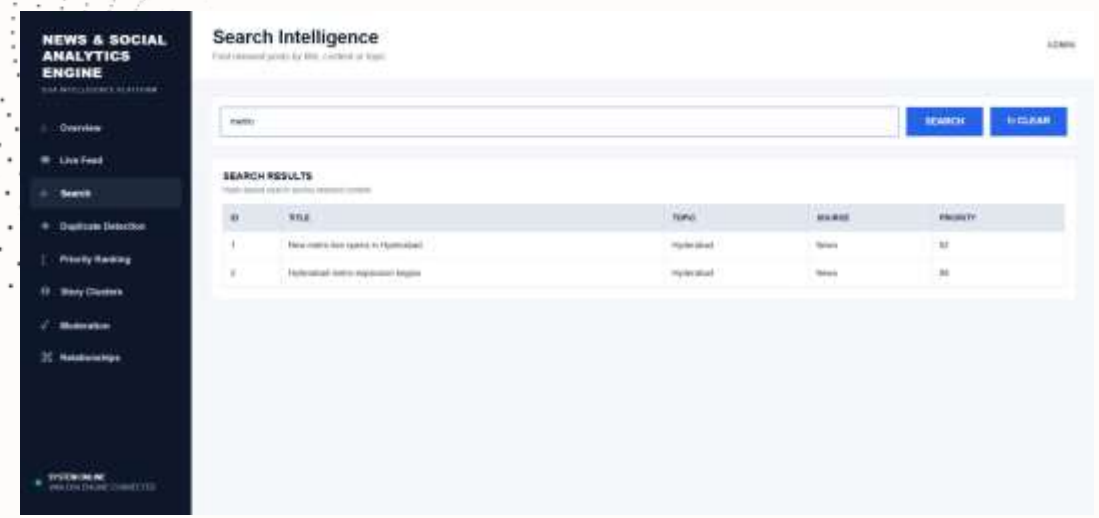
# Results

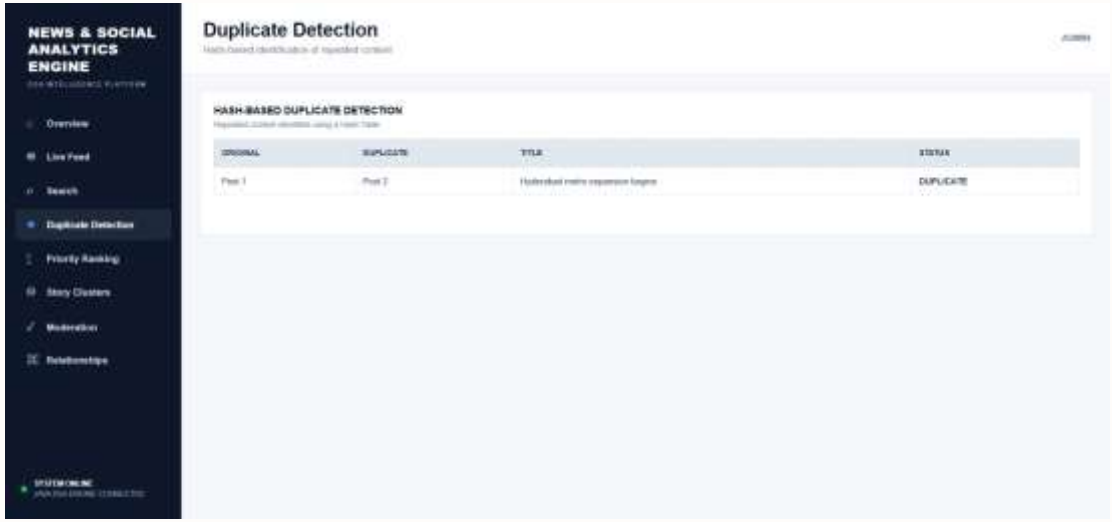

DATA STRUCTURES & ALGORITHMS
NEWS & SOCIAL MEDIA ANALYTICS ENGINE
Search. Analyze. Rank. Detect. Connect.
An intelligent platform for organizing and analyzing news and social media content using advanced data structures
LET'S START

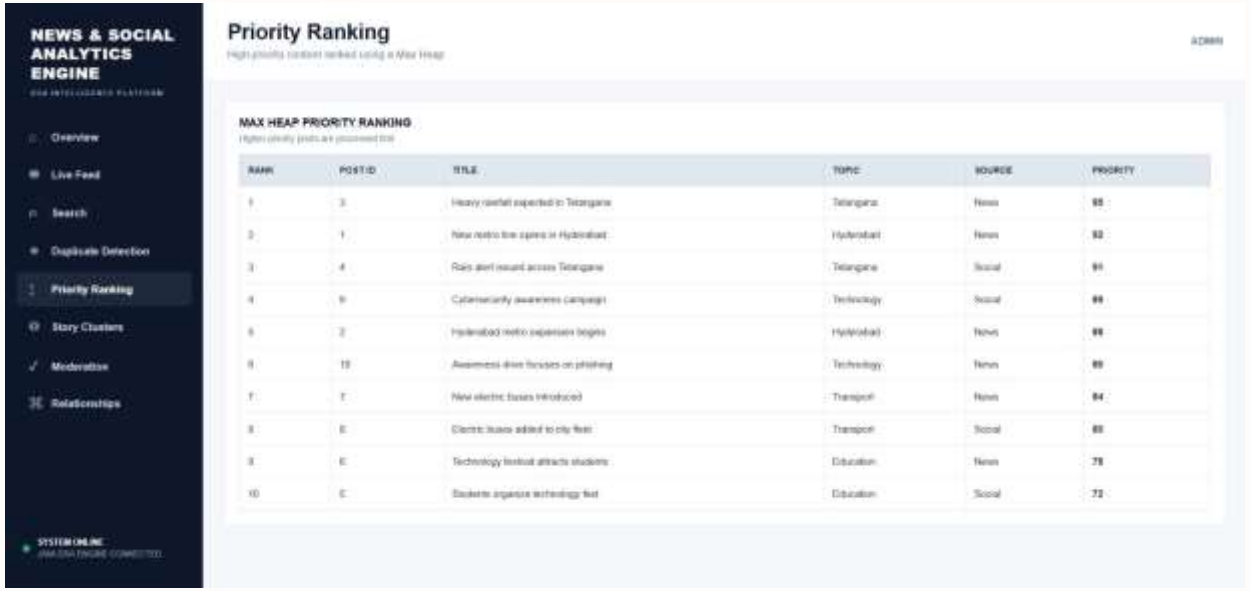

NEWS & SOCIAL ANALYTICS ENGINE
Overview
Live Feed
Search
Duplicate Detection
Priority Ranking
Story Clusters
Moderation
Relationships
Overview
ADMIN
TOTAL POSTS
10
TOPICS
5
DUPLICATES
1
MODERATION
10
PRIORITY STORIES
Heavy rainfall expected in Telangana
New metro line opens in Hyderabad
Rain alert issued across Telangana
Cybersecurity awareness campaign
Hyderabad metro expansion begins
TOPIC ACTIVITY
Education
Hyderabad
Technology
Telangana
Transport


NEWS & SOCIAL ANALYTICS ENGINE
Overview
Live Feed
Search
Duplicate Detection
Priority Ranking
Story Clusters
Moderation
Relationships
Live Feed
ADMIN
CONTENT FEED
ID
TITLE
TOPIC
SOURCE
PRIORITY

# Results

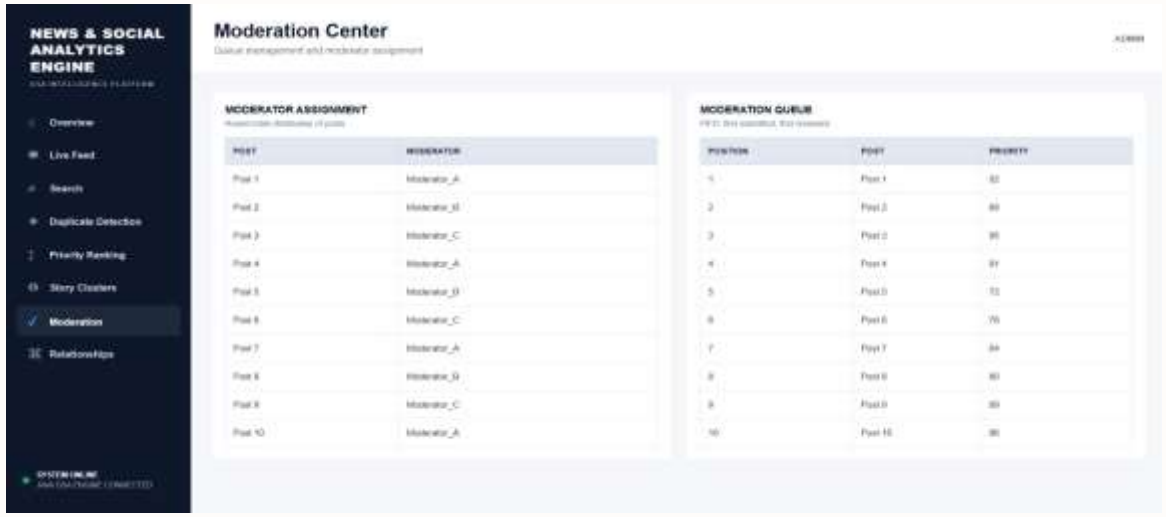

NEWS & SOCIAL ANALYTICS ENGINE
Search Intelligence
SEARCH RESULTS

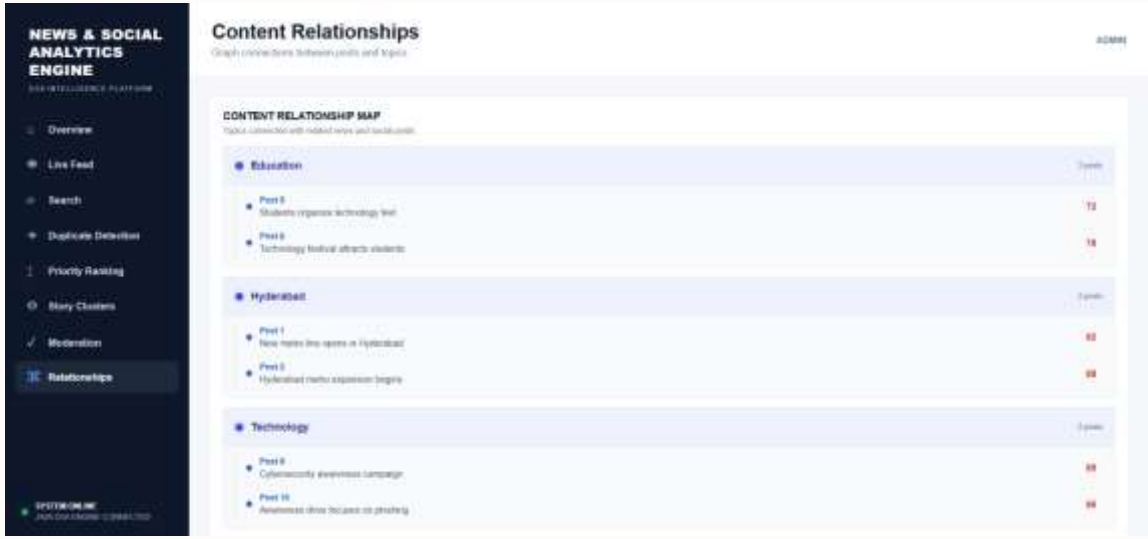

NEWS & SOCIAL ANALYTICS ENGINE
Duplicate Detection
HASH-BASED DUPLICATE DETECTION


NEWS & SOCIAL ANALYTICS ENGINE
Priority Ranking
MAX HEAP PRIORITY RANKING

# Results


NEWS & SOCIAL ANALYTICS ENGINE
Overview
Live Feed
Search
Duplicate Detection
Priority Ranking
Story Clusters
Moderation
Relationships
Story Clusters
RELATED STORY CLUSTERS
Education
Hyderabad
Technology
Telangana
Transport


NEWS & SOCIAL ANALYTICS ENGINE
Overview
Live Feed
Search
Duplicate Detection
Priority Ranking
Story Clusters
Moderation
Relationships
Moderation Center
MODERATOR ASSIGNMENT
POST
MODERATOR
MODERATION QUEUE
POSITION
POST
PRIORITY


NEWS & SOCIAL ANALYTICS ENGINE
Overview
Live Feed
Search
Duplicate Detection
Priority Ranking
Story Clusters
Moderation
Relationships
Content Relationships
CONTENT RELATIONSHIP MAP
Education
Hyderabad
Technology

# Hardware and Software requirements

**Hardware Requirements**

- Laptop or desktop computer
- Intel Core i3 or equivalent processor and above
- Minimum 4 GB RAM; 8 GB recommended
- Sufficient storage for source code and development tools

**Software Requirements**

- Operating System: Windows
- Programming Language: Java
- Frontend: HTML, CSS, JavaScript
- Server: Node.js and Express.js
- Input Data: CSV file
- Development Tool: Visual Studio Code

# Conclusion

The News and Social Media Analytics Engine demonstrates how Data Structures and Algorithms can be applied to organize and analyze text-based content. The project combines a Java processing engine with a web-based dashboard using Node.js and Express. Its features demonstrate searching, duplicate detection, priority ranking, story grouping, moderation, and relationship visualization. The working prototype helps connect theoretical DSA concepts with a practical application using sample data.

# Future Scope

- Integrate authorized live news APIs.
- Improve duplicate detection using advanced text-similarity techniques.
- Store posts and results in a database.
- Add filters for topics, dates, and sources.
- Improve moderation with user roles and authentication.
- Test performance with larger datasets.
- Improve error handling and automated testing.

(DEEMED TO BE UNIVERSITY)

# Thank you